

Tri-Modal Sequence Model (TSM): Research-Oriented Hybrid AI Architecture

Research Focus

Investigates limits of Transformer attention ($O(n^2)$) and develops sub-quadratic hybrid architecture using attention, state-space, and convolution.

Project Objective

Design a scalable, efficient sequence model combining multiple memory paradigms.

Core Idea

Transformer (attention), Mamba (state-space), Hyena (convolution) form a tri-modal memory system.

Architecture

Input → Embedding → Parallel (Mamba | Transformer | Hyena) → Fusion → Feedforward → Output

Adaptive Gating

Dynamic weighting of modules to route information effectively.

Hierarchical Memory

Local (Transformer), Global (Hyena), Long-term (Mamba) memory layers.

Context Scaling

Evaluate across sequence lengths (512, 2K, 8K).

Operator Framework

$F(x) = \sum \alpha_i O_i(x)$ representing mixture of operators.

Ablation Studies

- 1 Transformer only
- 2 Mamba only
- 3 Hyena only
- 4 Hybrid comparison

Challenges

- 1 Training instability
- 2 Representation mismatch
- 3 Compute complexity

Expected Outcomes

- 1 Working prototype

- 2 Improved long-context efficiency
- 3 Insights into hybrid memory systems
- 4 Potential publication